ΚΑΤΟΙΚΩΝ ΕΡΓΑΣΙΑ 1

1.1

② β) $f(x) = 2x^3 - x$

$$g(x) = \begin{cases} \frac{1}{x-1}, & x > 1 \\ 2x, & x \le 1 \end{cases}$$

$g(0) = 0$

$g(1) = 2$

$g(-1) = -2$

$g(\sqrt{3}) = \frac{1}{\sqrt{3}-1}$

$g(\frac{1}{3}) = \frac{2}{3}$

$$g(a) = \begin{cases} \frac{1}{a-1}, & a > 1 \\ 2a, & a \le 1 \end{cases}$$

$$g(\sqrt{a}) = \begin{cases} \frac{1}{\sqrt{a}-1}, & a > 1 \\ 2\sqrt{a}, & 0 \le a \le 1 \end{cases}$$

$$g(\tfrac{1}{a}) = \begin{cases} \frac{1}{\frac{1}{a}-1}, & a > 1 \\ \frac{2}{a}, & 0 < a \le 1 \end{cases}$$

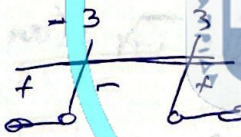
$$g(a^2) = \begin{cases} \frac{1}{a^2-1}, & a > 1 \\ 2a^2, & a \le 1 \end{cases}$$

③ α) $f(x) = 2x^3 - x$, $x \in \mathbb{R}$.

(β) $x^2 - 9 \ge 0$

$\Leftrightarrow x^2 \ge 9$

$\Leftrightarrow x \ge \pm 3$



$A_f = x \in (-\infty, 3] \cup [3, +\infty)$

(γ) $x \in \mathbb{R}$, $x^2 + 9 \ge 0$

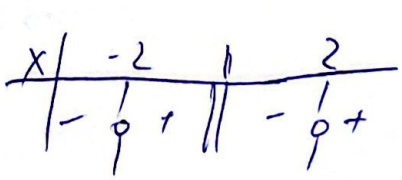
(δ) $2x - 3 \ne 0$

$x \ne \frac{3}{2} \rightarrow A_f = x \in \mathbb{R} - \{\frac{3}{2}\}$.

(ε) $x \ne \frac{1}{2} \rightarrow A_f = x \in \mathbb{R}\{\frac{1}{2}\}$.

(στ) $\frac{x^2-4}{x-1} \ge 0 \Rightarrow$ $x = 2$, $x = -2$, $x \ne 1$

$x \ne 1$



$A_f \Rightarrow x \in [-2, 1) \cup [2, +\infty)$.

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(ζ) $x \in \mathbb{R}$.

(η) $x \geq 0$, $x \in [0, +\infty)$.

(θ) $x \geq 0$, $x \in [0, +\infty)$

(ι) $\sin x \geq 0$ $x \in [0, \pi] \cup [\pi,$

(ια) $2 - \cos x \neq 0$

$\cos x \neq 2$. $x \in \mathbb{R}$.

(ιβ) $\cos x \neq 1$

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(ιγ) α) $y \in \mathbb{R}$.

(β) $y^2 = x^2 - 9$ $y \in [0, +\infty)$

$x = -\sqrt{y^2 + 9}$

(ε) $y = \frac{2x^2 - 1}{2x - 1} \Rightarrow 2xy - y = 2x^2 - 1 \Rightarrow 2xy - 2x^2 = y - 1$

$\Rightarrow x(2y - 2x) = y - 1$

$x = \frac{y - 1}{2y - 2x}$

(ιη)

1.2 $A_f \in \mathbb{R}$ $A_g \in [0,+\infty)$.

① α)(i) $(f+g)(x) = 2x + \sqrt{x}$ $A_{f+g} [0,+\infty)$

(ii) $(f-g)(x) = 2x - \sqrt{x}$ $A_{f-g} [0,+\infty)$.

(iii) $(fg)(x) = 2x\sqrt{x}$ $A_{fg} [0,+\infty)$

(iv) $\left(\frac{f}{g}\right)(x) = \frac{2x}{\sqrt{x}}$ $A_{\frac{f}{g}} (0,+\infty)$

(v) $\left(\frac{g}{f}\right)(x) = \frac{\sqrt{x}}{2x}$ $A_{\frac{g}{f}} (0,+\infty)$

② γ) $A_f \in \mathbb{R}$. $A_g \in [1,+\infty)$

(i) $(fog)(x) = f(g(x)) = \left(\sqrt{x-1}\right)^2 = x-1$

$A_{fog} = \{x \in [1,+\infty) : \sqrt{x-1} \in \mathbb{R}\} = [1,+\infty)$

(ii) $(fof)(x) = f(f(x)) = (x^2)^2 = x^4$

$A_{fof} = \{x \in \mathbb{R} : x^2 \in \mathbb{R}\} = \mathbb{R}$

(iii) $(gof)(x) = g(f(x)) = \sqrt{x^2-1}$

$A_{gof} = \{x \in \mathbb{R} : x^2 \in [1,+\infty)\} = \mathbb{R} \cdot [1,+\infty)$

(iv) $(gog)(x) = g(g(x)) = \sqrt{\sqrt{x-1}-1}$

$A_{gog} = \{x \in [1,+\infty) : \sqrt{x-1} \in [1,+\infty)\}$

$x \geq 1$, $\sqrt{x-1} \geq 1$, $x-1 \geq 1$, $x \geq 2$ $\Rightarrow A_{gog} \in [2,+\infty)$.

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③α) i) a) $\log\circ h$.

$$\log = f(g(x)) = \frac{1}{x^2-1}$$

$$\log(h(x)) = \frac{1}{x^6-1}$$

$$\log\circ h = \{x \in \mathbb{R} : x^3 \in \mathbb{R}-\{\pm 1\}\} = x \in \mathbb{R}-\{\pm 1\}.$$

ii) $h\circ\log$

$$h\circ f = \frac{1}{x^3}$$

$$h\circ\log = \frac{1}{(x^2-1)^3} \qquad x \in \mathbb{R}-\{\pm 1\}.$$

iii) $h\circ\log$ και $h\circ h = \frac{1}{x^3}$.

$$h\circ\log = \frac{1}{(x^2-1)^3} \qquad x \in \mathbb{R}-\{\pm 1\}.$$

iv) $\log\circ f$

$$\log = \frac{1}{x^2-1}$$

$$\log\circ f = \frac{1}{\frac{1}{x^2}-1} \qquad x \in \mathbb{R}-\{\pm 1, 0\}.$$

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(4) β) $f_2(x) = \sqrt{\frac{1}{\ln(x^2+1)}}$

$h(x) = \sqrt{\frac{1}{\ln(x)}}$ $\quad g(x) = x^2+1$

$h \circ g = \sqrt{\frac{1}{\ln(x^2+1)}}$

(6) γ) i) $f(x) = x-1$

$1-1$

ii) $f(x) = x^2-1$

Δεν είναι 1-1

(7) α) $f(g(x)) = 4x+1$ } Δεν είναι αντίστροφες

$g(f(x)) = 4x-3$ }

γ) $f(g(x)) = \frac{\frac{x-2}{x-1} - 1}{\frac{x-2}{x-1} - 2} = \frac{1}{x}$

$g(f(x)) = \frac{\frac{x-1}{x-2} - 2}{\frac{x-1}{x-2} - 1} = -x+3$

} Δεν είναι αντίστροφες

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⑧ α) $f(x) = 3x-4$

$y = 3x-4$

$y = \frac{1}{3}x + \frac{4}{3}$

$f^{-1}(x) = \frac{1}{3}x + \frac{4}{3}$

β) Δεν υπάρχει

γ) $y = 3x^3+9$

$f^{-1}(x) = \sqrt[3]{\frac{1}{3}x - 3}$

δ) $y = \frac{x-1}{x-2}$

$f^{-1}(x) = \frac{-2x+1}{1-x}$

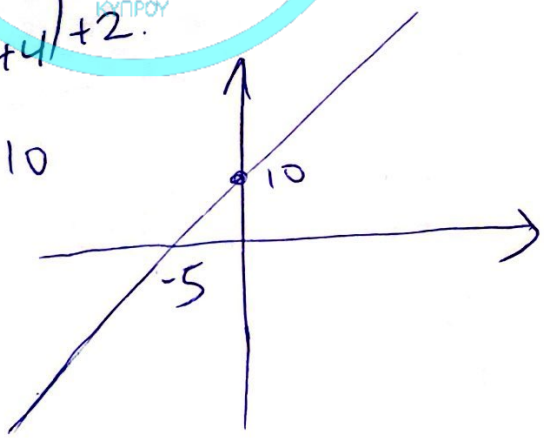
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1.3 ① β)

$x = t-4$

$t = x+4 \Rightarrow y = 2(x+4)+2$

$y = 2x+10$



6.

α.) $x = t + 2$

$\boxed{t = x - 2}$

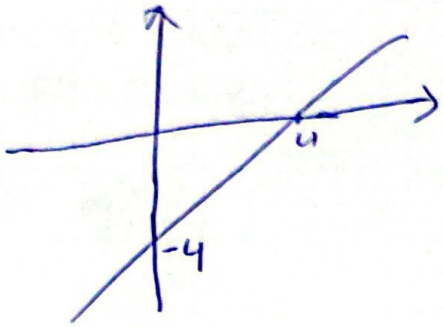
$\Rightarrow y = x - 2 - 2$

$\Rightarrow y = x - 4$

Q.

α,

β,



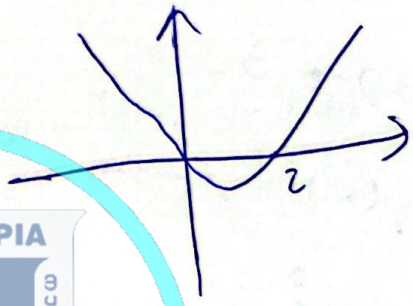
ε)

$x = 2t + 1 \Rightarrow \boxed{t = \frac{x-1}{2}}$

$y = 4t^2 - 1$

$\Rightarrow y = 4\left(\frac{x-1}{2}\right)^2 - 1 = (x-1)^2 - 1$

Z



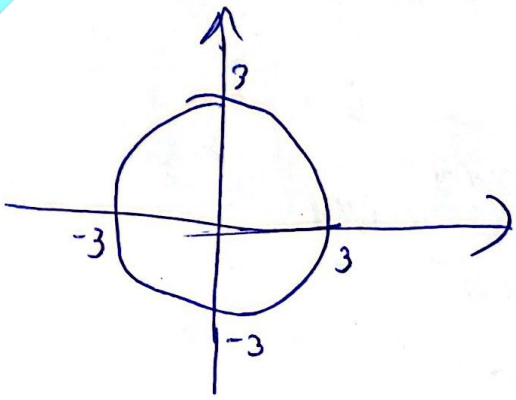
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δ) $x = 3\sin(\pi t)$

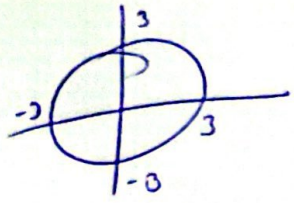
$y = 3\cos(\pi t)$

$x^2 + y^2 = 9\sin^2(\pi t) + 9\cos^2(\pi t) = 9\underbrace{\left(\sin^2(\pi t) + \cos^2(\pi t)\right)}_{1} = 9$

$\boxed{x^2 + y^2 = 9}$



② α)



$x = 3\sin t$

$y = 3\cos t$

β)

2.1

① α) $\lim_{x \to 0^{\pm}} g(x) = 3.$

$\lim_{x \to 0} g(x) = 3$

$\lim_{x \to -\infty} g(x) = \nexists$

$g(0) = 3$

ε) $\lim_{x \to 0^-} f(x) = 1$ $\lim_{x \to 0^+} f(x)$ $\lim_{x \to 0} f(x)$ $\lim_{x \to +\infty} f(x)$ $f(0) = -2$

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στ) $\lim_{x \to 4} G(x) = 1$

$\lim_{x \to 0^-} G(x) = 3$

$\lim_{x \to 0^+} G(x) = 3$

$\lim_{x \to +\infty} f(x) = 0$

$\lim_{x \to -\infty} G(x) = \nexists$

2.2

6.2

① α) $f(x) = 3x$

$L = 9$

$\varepsilon = 0,1$

$a = 3$

$0 < |x-3| < \delta$.

$|3x - 9| < 0,1$

β)

② α) $|1-2x-7| < \varepsilon$ $\quad 0 < |x+3| < \delta$

$|-2x-6| < \varepsilon$

$-2|x+3| < \varepsilon$

$|x+3| < \frac{\varepsilon}{2}$.

άρα $\lim_{x \to -3} (1-2x) = 7$

③ α) $\left|\frac{2x}{x+1} - 2\right| < \varepsilon$

$2x < (\varepsilon - 2)(x+1)$

$2x < \varepsilon x + \varepsilon - 2x - 2$

$\frac{2-\varepsilon}{\varepsilon} < x$ $\quad N = \frac{2-\varepsilon}{\varepsilon}$ άρα $\lim_{x \to \infty} \frac{2x}{x+1} = 2$

④ α) $\lim_{x \to 1} \frac{2}{(x-1)^2} = \infty$

$\frac{1}{(x-1)^2} > M$.

$(x-1)^2 < \frac{1}{M}$.

$-\frac{1}{\sqrt{M}} < x-1 < \frac{1}{\sqrt{M}} \Rightarrow \delta = \frac{1}{\sqrt{M}}$.

Άρα $\lim_{x \to 1} \frac{1}{(x-1)^2} = +\infty$.

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$|x| \geq 1$

ε) ⑦ α) $\lim_{x \to a} [1+2] = 3.$

στ) β) $\lim_{x \to a} [1+2 \cdot 0] = 1$

δ) $\lim_{x \to a} \frac{0}{-2} = 0$

δ) $\lim_{x \to a} \frac{2 \cdot 1 + (-2)}{0} = \frac{0}{0}$ (Απ.).

ζ) ε) $\lim_{x \to a} [1 (-2)]^2 = 4$

ζτ) $\lim_{x \to a} \sqrt{1 \cdot 2} = \sqrt{2}$

η) ⑧ α) $\lim_{x \to 1+} (x^3 - 2x^2 + 2x - 1) = 0$

β) $\lim_{x \to -1-} (-2x^4 - 4x^3 - 2x^2) = 0$

γ) $\lim_{x \to 3} [(x-2)(x-1)(x+1)^2] = 32.$

θ δ) $\lim_{x \to 100} (2x^2 - 3x^3 + 4x^{-4} + x) = \lim_{x \to \infty} -3x^3 = -\infty$

βτ) $\lim_{x \to -2} \sqrt{y^3 + 5} = (\text{Απ.})$

ι) $\lim_{x \to 1+} \frac{1}{(x-1)^3} = +\infty$

ιβ) $\lim_{x \to -1} \frac{x^2 + x}{x - 1} = \frac{0}{-2} = 0$

ιη) $\lim_{x \to \infty} \frac{x^2 - 3x + 2}{x - 1} = \lim_{x \to +\infty} \frac{x^2}{x} = +\infty$

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2.3

② β) $f(x) = \sqrt{1-x^2}$

$x^2 \geq 1$

$|x| \geq 1$

ε) $f(x) = \dfrac{x+1}{x^2+2x+3}$ Δεν υπάρχει ασύμπτωτα.

6ζ) $f(x) = \dfrac{x+1}{x^2+3x+2}$

$x^2+3x+2 \neq 0$.

$(x+2)(x+1) \neq 0$.

$x \neq -2$ $|x \neq -1$

1α) $f(x) = \begin{cases} \dfrac{x^2+1}{x-1}, & x>1 \\ x+1, & x \leq 1 \end{cases}$ $\longrightarrow$ $x \neq 1$

④ δ) $f(x) = \dfrac{x+1}{|x|+1}$

$\dfrac{x+1}{x+1} = 1$, $x \geq 0$

$\dfrac{x+1}{-x+1}$, $x < 0$

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⑤ α) $x^3+x-5=0$, $[0,2]$.

$f(0) = -5$

$f(2) = 5$

$\Big\} f(0) \cdot f(2) < 0 \Rightarrow$ Βολζάνο $\exists$ τουλάχιστον μία λύση στο διάστημα $[0,2]$

3.2

2.4

(3) δ) $\lim_{x\to 0} \frac{\sin x}{\cos x - 1} = \frac{0}{0}$ (Απρ).

DLH $\Rightarrow \lim_{x\to 0} \frac{\sin x}{\cos x - 1} = \lim_{x\to 0} \frac{\cos x}{-\sin x} = \infty$.

η) $\lim_{x\to 0} \frac{\tan 5x}{\sin 3x} = \frac{0}{0}$ (Απρ).

DLH $\lim_{x\to 0} \frac{5\tau\varepsilon\varphi^2 5x}{3\cos 3x} = \frac{5 \cdot \frac{1}{\cos^2 5x}}{3 \cdot \cos^3 x} = \frac{5}{3}$.

(5) β) $\lim_{x\to \pi} \frac{x - \pi}{\sin 2x}$

DLH $\Rightarrow \lim_{x\to \pi} \frac{1}{2\cos 2x} = \frac{1}{2}$.

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3.1

(1) β) $\lim_{x\to -2} \frac{f(x) - f(-2)}{x+2} = \lim_{x\to -2} \frac{\frac{1}{x+1} + 1}{x+2} = \lim_{x\to -2} \frac{1+x+1}{(x+1)(x+2)}$

$= \lim_{x\to -2} \frac{x+2}{(x+1)(x+2)} = -1$

3.2

6.: ① β) $y' = \frac{1}{2\sqrt{x}} - \frac{3\cdot 5}{2\sqrt{5x}} + \frac{3x^7}{2\sqrt{x^3}} + 4\frac{1}{\sqrt{x}}$

③

a) γ) $y' = \frac{-1}{x^2} + \frac{3}{x^2} - \frac{6}{9x^2} + \frac{3}{9}$

β)

ε) $y' = \frac{2x}{2\sqrt{x^2+5}}$

η) $y' = 6x\,e^{3x^2}$

⑦ ς) $y' = 2x\sin x + x^2\cos x$

ζ) $y' = \sin\sqrt{x} + x\cdot\frac{1}{2\sqrt{x}}\cdot\cos\sqrt{x}$

η) $y' = \frac{2x\sin x - x^2\cos x}{\sin^2 x}$

6. ΚΕΦ) $y' = \frac{\left(-\frac{3}{2}\sin 3x\right)(1-3x) + \frac{3}{2}\cos 3x}{(1-3x)^2}$.

(

② β) $\frac{x'}{2\sqrt{2x+1}} = \frac{1}{2\sqrt{2x+1}}$

2.

δ) $\frac{\frac{1}{2\sqrt{x}}}{\frac{1}{3\sqrt{x}}} = \frac{1}{\sqrt{x}}$

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④ (2) α) $\frac{d^2y}{dx^2} = \left(-2 - \frac{3\frac{dy}{dx}}{dx}\right)(3x+2y) - (-2x-3y)\left(3+2\frac{dy}{dx}\right)$

③ α) $\frac{dw}{dx} = \frac{f'(x)}{2\sqrt{f(x)}}$

δ) $\frac{dw}{dx} = \frac{f'(x)}{f(x)}$.

3.3

① δ) $y'' = \frac{x \cdot 2\sqrt{x^2+5} - 2x \cdot \frac{4x}{4\sqrt{x^2+5}}}{(2\cdot\sqrt{x^2+5})^2}$

ε) $y(5) = \frac{1}{2}e^x + 24e^{2x} + 162e^{-3x}$

3.4

① α) $2x + 3y + 3x\frac{dy}{dx} + 2y\frac{dy}{dx} = 0$.

$\frac{dy}{dx} = \frac{-2x-3y}{3x+2y}$.

γ) $3x^2y^3 + x^3 3y^2\frac{dy}{dx} - 4x - 4x\frac{dy}{dx} = 0$.

$\frac{dy}{dx} = \frac{4x - 3x^2y^3}{x^3 3y^2 - 4x}$.

ε) $\sin y + x\cos y \cdot \frac{dy}{dx} + \frac{dy}{dx}\sin x + y\cos x = 1 + \frac{dy}{dx}$.

$\frac{dy}{dx} = \frac{1 - \sin y - y\cos x}{x\cos y + \sin y - 1}$

(14)

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④ (2α) $\frac{d^2y}{dx^2} = \frac{(-2-3\frac{dy}{dx})(3+2\frac{dy}{dx}) - (-2x-3)(3+2\frac{dy}{dx})}{(3x+2y)^2}$.

3.5

η' (2α) $dy = 3t^2$

$dx = 2t$ $\quad \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{3t^2}{2t} = \frac{3}{2}\cdot t$.

(6) $dy = 3\cos\theta$.

$dx = -2\sin\theta$ $\quad \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{3\cos\theta}{-2\sin\theta}$

③ $dy = 2x - 4$

$dx = \frac{1}{2}(2t^2+1)^{\frac{3}{2}} \cdot (4t) = 2t(2t^2+1)^{\frac{2}{2}}$.

$\frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{2x-4}{2\cdot\sqrt{2}(2\cdot 2+1)^{\frac{3}{2}}}$

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4.1

① α) $f(x) = x^2 - 4x + 5$

$f'(x) = 2x - 4$

$f'(x) = 0 \Rightarrow \boxed{x = 2}$

x		2	
	−	0	+
	↘	↓	↗

f φθίνουσα στο $(-\infty, 2]$.
f αύξουσα στο $[2, +\infty)$.

③ γ) $y = 4x^2 + 12x + 6$

$y' = 12x^2 + 12$.

$y' = 0 \Rightarrow 12x^2 = -12$.

$\Rightarrow x^2 = -1$ Αδύνατη. Δεν υπάρχουν Τ.Α.

ε) $y = x^{\frac{4}{3}}(1-x)^{\frac{1}{3}}$

$y' = \frac{4\sqrt[3]{x - x^2}}{3} - \frac{x\sqrt[3]{x}}{3\sqrt[3]{(1-x)^2}}$.

$y' = 0$

θ) $y = \frac{x^2}{x^2+1}$

$y' = \frac{2x}{(x^2+1)^2}$ $y' = 0 \Rightarrow \boxed{x = 0}$

x		0	
	−	0	+
	↘	↓	↗

Φθίνουσα στο $(-\infty, 0]$.
Αύξουσα στο $[0, +\infty)$.

ιβ) $y = \cos^2 x$

$y' = -2\eta\mu x$

$y' = 0 \Rightarrow \boxed{x = k\pi}$

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④ γ) $y' = 4x^3 - 4x$

$y'' = 12x^2 - 4$

$y'' = 0 \Rightarrow 12x^2 = 4$

$x^2 = \frac{1}{3}$

$x = \pm \frac{1}{\sqrt{3}}$

x	$-\frac{1}{\sqrt{3}}$		$\frac{1}{\sqrt{3}}$	
	+	0	−	0	+
	↗		↘		↗

η) $y'' = \frac{12}{(x-3)^3}$

$x \neq 3$

x	3	
	+	+
	↗	↗

θ) $y'' = \frac{-6x^2 + 2}{(x^2+1)^3}$

$y'' = 0 \Rightarrow 6x^2 = 2$

$x^2 = \frac{1}{3}$

$x = \pm \frac{1}{\sqrt{3}}$

x	$-\frac{1}{\sqrt{3}}$		$\frac{1}{\sqrt{3}}$	
	+	0	−	0	+
	↗		↘		↗

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⑥ 6) $f'(x) = 6x^2 + 6x - 12.$

$f'(x) = 0 \Rightarrow x^2 + x - 2 = 0.$

$(x+2)(x-1) = 0$

$x = -2 \quad x = 1$

x	-2		1	
	+	0	−	0	+
	↗	max	↘	min	↗

$f(-3) = 9$

$f(3) = 45$ Max (3, 45)

$f(-2) = 20$

$f(1) = -7$ min (1, -7)

(γ) $f(x) = \frac{\ln x}{2x}$

$f'(x) = \frac{1 - \ln x}{2x^2}$ $\quad f'(x) = 0$

$(x \neq 0)$ $\quad \ln x = 1$

$x = e$

x	0	e
f	∄	+ 0 −
	↗	↗ max ↘

$f(1) = 0$

$f(e) = \frac{1}{2e}$

Δ α) $f'(x) = -4 - 8x$

$f'(x) = 0 \Rightarrow 8x = -4$

$x = -\frac{1}{2}$

x	$-\frac{1}{2}$
f	+ 0 −
	↗ max ↘

$f(-\frac{1}{2}) = 7$ $\quad (-\frac{1}{2}, 7)$

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δ) $f(x) = \frac{x+1}{x-1}$

$f'(x) = \frac{-2}{(x-1)^2}$ $\quad f'(x) = 0$

x	1
f	− ∄ −
	↘ ∄ ↘

$f(-1) = 0$ $\quad (-1, 0)$

$f(1) = \not\exists$

δ) $f(x) = \frac{2}{4-x^2}$

6.

$f'(x) = \frac{4x}{(4-x^2)^2}$

③

α) $x = 0$

$x = \pm\sqrt{2}$

x	$-\sqrt{2}$	0	$\sqrt{2}$	
	− ‖ −	0 +	‖ +	
	↘	↗	↗	

β) $f(0) = \frac{1}{2}$ $\quad (0, \frac{1}{2})$.

⑩

3,5 / 3,428571428

4.2

⑦

① α) $f(x) = x^2 - 4x + 5$

$f'(x) = 2x - 4$

$f''(x) = 2.$ Συνεχής και για παντού.

② α) $f(x)' = 12x^3 - 12x^2 = 0 \Rightarrow$ $x = 0$ $x = 1$

x	0	1	
	+ 0	− 0	+
	↗	↘	↗

6

(1

$f''(x) = 36x^2 - 24x = 0 \Rightarrow$ $x = 0$ $x = \frac{2}{3}$

x	0	$\frac{2}{3}$	
	+ 0	− 0	+
	↗	↘	↗
	Σκ		Σκ

$f(0) = 0$

$f(1) = -1$

$f(\frac{2}{3}) = \frac{4}{27}$

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4.5

β) $f(x) = \frac{x^2}{x^2+2}$ $f'(x) = \frac{4x}{(x^2+2)^2} = 0 \Rightarrow x=0$

x		0	
	−	0	+
	↓	y	↑

$f''(x) = \frac{-12x^2+8}{(x^2+2)^3} = 0 \Rightarrow x = \pm\frac{\sqrt{6}}{3}$

x		$-\frac{\sqrt{6}}{3}$		$\frac{\sqrt{6}}{3}$	
	−	0	+	0	−
	↓		↑		↓

$f(0) = \frac{1}{2}$

$f\left(-\frac{\sqrt{6}}{3}\right) = \frac{1}{4}$

$f\left(\frac{\sqrt{6}}{3}\right) = \frac{1}{4}$

γ) $f'(x) = 2x\ln x + x = 0 \Rightarrow x = 0$

$x = \frac{\sqrt{e}}{e}$

$f''(x) = 2\ln x + 3 = 0 \Rightarrow x = e^{-\frac{3}{2}}$

$x = \frac{\sqrt{e}}{e^2}$

$f\left(\frac{\sqrt{e}}{e}\right) = -0{,}184$

x		$\frac{\sqrt{e}}{e}$	
	−	0	+
	↓		↑

x		$\frac{\sqrt{e}}{e^2}$	
	−	0	+
	↓		↑

$f\left(\frac{\sqrt{e}}{e^2}\right) = -0{,}075.$

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4

β) $S_{10} = i$ $d = 2$.

$a_1 = 0$

$a_{10} = 18$

$S_{10} = \left(\frac{18}{2}\right) \cdot 10 = 90$

β) $S_0 = \frac{\frac{19683}{512}(1-1{,}5^{10})}{1-1{,}5} = 4356{,}78$ $\quad a_{10} = \beta \cdot 1{,}5^9 = \frac{19683}{512}$

γ) $S_{100} = \frac{1{,}26\cdot10^{-69}(2-0{,}2^{100})}{1-0{,}2} \simeq 3{,}15\cdot10^{-69}$ $\quad a_{100} = 2\cdot 0{,}2^{99} = 1{,}26\cdot10^{-69}$

(20)

4.5

① ε) $\sum_{n=2}^{\infty} \frac{n^2}{2^n(n-1)}$.

4.7

① α) $f(x) = \cos x$

$f(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots$

② β) $f(x) = \ln(2x-1)$.

$f(x) = \frac{2(x-1)}{1!} - \frac{4(x-1)^2}{2!} + \frac{16(x-1)^3}{3!} + \cdots$

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5.1

6. ① β) $\int (7x^{7/4} - 5x^{5/2} + 3x^{3/2} - x^{1/2})\,dx.$

③ α) $= \frac{14x^4\sqrt{x}}{9} - \frac{10x^3\sqrt{x}}{7} + \frac{6x^2\sqrt{x}}{5} - \frac{2x\sqrt{x}}{3} + C.$

β 5) $\int \frac{x^3 - x + 3}{2x^3}\,dx = \frac{x}{2} + \frac{1}{2x} - \frac{3}{4x^2} + C$

η) $\int \sqrt{3x-1}\,dx = \frac{2\sqrt{3x-1}\,(3x-1)}{9} + C$

θ) $\int \frac{1}{(2-y)^3}\,dy = \frac{1}{2(2-y)^2} + C$

ιβ) $\int \frac{x+3}{1-2x}\,dx = -\frac{x}{2} - \frac{7}{4}x\ln x(|1-2x|) + C$

κ) $\int \sec 3x \tan 3x\,dx = \frac{1}{3}\sec 3x + C$

5.2

① β) $I = \frac{(t^3+t-2)^{\frac{7}{4}}}{7} + C$

δ) $I = 3\int \frac{t}{(t^2+3)^{\frac{1}{3}}}\,dt = \frac{3}{2}\int 2t(t^2+3)^{-\frac{1}{3}}\,dt$

$= \frac{\frac{3}{2}(t^2+3)^{\frac{2}{3}}}{\frac{2}{3}} + C.$

ι) $I = \int \frac{x}{\sqrt{x^2+1}}\,dx = \frac{1}{2}\int 2x(x^2+1)^{-\frac{1}{2}}\,dx = \frac{1}{2}\frac{(x^2+1)^{\frac{1}{2}}}{\frac{1}{2}}$

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16ε)

2α) $I=\int \frac{1}{\sqrt{12+4x-x^2}}dx = \int \frac{1}{\sqrt{-(x^2-4x-12)}}dx = \int \frac{1}{\sqrt{-(x^2-4x+4-16)}}dx$

$= \int \frac{1}{\sqrt{[-(x-2)^2-16]}}dx = \int \frac{1}{\sqrt{16-t^2}}dt = \arcsin\left(\frac{t}{4}\right)$

$= \arcsin\left(\frac{x-2}{4}\right)+c$

2γ) $t=x-2$

$dt=dx$

$x=t+2$

5ιη) $I=\int \frac{x+1}{x^2-4x+8}dx = \int \frac{\frac{1}{2}(2x-4)}{x^2-4x+8} + \frac{3}{x^2-4x+8}dx$

(1

$= \frac{1}{2}\ln\left(|x^2-4x+6|\right) + \frac{\arctan\left(\frac{x-2}{2}\right)}{2}+c$

2

κθ) $I=\int \frac{\cos\sqrt{x}}{2\sqrt{x}}dx = \int \cos t\,dt = \sin t = \sin\sqrt{x}+c$

$t=\sqrt{x} \Rightarrow x=t^2$

$dt=\frac{1}{2\sqrt{x}}dx$

$dx=2\sqrt{x}\cdot dt$

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ε) $I=\int \sin^4 x\cos x\,dx$

$= 5\int \sin^4 x\cos x\,dx$

$= 5\int t^4\,dt$

$= \frac{5t^5}{5} = \sin^5 x + c$

$t=\sin x$

$dt=\cos x\,dx$

$dx=\frac{dt}{\cos x}$

2α) $I = \int e^{\cos x} \sin x\, dx$ $\quad t = \cos x$

$= -\int e^t dt$ $\quad dt = -\sin x\, dx$

$= -e^{\cos x} + C$ $\quad dx = \frac{dt}{-\sin x}$

2γ) $I = \int (e^x+1)^2 e^x dx$

$= \frac{(e^x+1)^3}{3} + C$

5.3

① γ) $I = \int \frac{x}{x^2-3x-4} dx = \frac{4}{5}\ln(|x-4|) + \frac{1}{5}\ln(|x+1|) + C$

$\frac{x}{x^2-3x-4} = \frac{x}{(x-4)(x+1)} = \frac{A}{x-4} + \frac{B}{x+1} = \frac{A(x+1)+B(x-4)}{5(x-4)} + \frac{1}{5(x+1)}$.

Για $x=-1$: $-1 = -5B$ $\Rightarrow \boxed{B = \frac{1}{5}}$

Για $x=4$: $4 = 5A$ $\Rightarrow \boxed{A = \frac{4}{5}}$

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η) $I = \int \frac{1+x}{(4+x^2)(x+2)} dx$

$\frac{1+x}{(x^2+4)(x+2)} = \frac{Ax+C}{x^2+4} + \frac{B}{x+2} = \frac{(Ax+C)(x+2)+B(x^2+4)}{x^2+4} = \frac{\left(\frac{x}{2}+\frac{3}{4}\right)}{x^2+4} - \frac{1}{8(x+2)}$

Για $x=-2$: $-1 = 8B$ $\Rightarrow \boxed{B = -\frac{1}{8}}$ Για $x=1$: $2 = \left(\frac{3}{4}+A\right)2 - \frac{1}{8}\cdot 4$ $\Rightarrow \boxed{A = \frac{1}{2}}$

Για $x=0$: $1 = 2C + 4B$. $\Rightarrow \boxed{C = \frac{3}{4}}$

γ) $I=\int \frac{2x}{(x-1)^2(x+2)}dx = \frac{4}{9}\int\frac{1}{x-1}dx + \frac{2}{3}\int(x-1)^{-2}dx - \frac{4}{9}\int(x-1)^{-2}dx = \frac{4}{9}\ln|x-1| + \frac{2}{3}\frac{(x-1)^{-1}}{-1} - \frac{4}{9}\frac{(x-1)^{-1}}{-1} + c$

$\frac{2x}{(x-1)^2(x+2)} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x+2} = \frac{4}{9(x-1)} + \frac{2}{3(x-1)^2} - \frac{4}{9(x-1)^2}$

$2x = A(x-1)(x+2) + B(x+2) + C(x-1)^2$.

Για $x=-2$: $-4 = 9C$

$\boxed{C=-\frac{4}{9}}$

Για $x=1$: $2 = 3B$

$\boxed{B=\frac{2}{3}}$

Για $x=-1$: $-2 = -2A + B + 4C$

$-2 = -2A + \frac{2}{3} - \frac{16}{9}$

$\boxed{A=\frac{4}{9}}$

5.4

① β) $I=\int x^2e^x dx = \int x^2 d(e^x) = x^2e^x - 2\int e^x \cdot x\,dx$.

$= x^2e^x - 2\int x\,d(e^x)$.

$= x^2e^x - 2\left[xe^x - \int e^x dx\right] = x^2e^x - 2xe^x + 2e^x + c$.

ε) $I=\int 2x\cos 2x\,dx = \int 2x\,d\left(\frac{\sin 2x}{2}\right) = x\sin 2x - \int \sin 2x\,dx$

$= x\sin 2x + \frac{\cos 2x}{2} + c$

η) $I=\int x^2\ln x\,dx = \int \ln x\,d\left(\frac{x^3}{3}\right) = \frac{x^3}{3}\ln x - \int \frac{x^3}{3x}dx = \frac{x^3}{3}\ln x - \frac{1}{3}\int x^2dx$.

$= \frac{x^3}{3}\ln x - \frac{x^3}{9} + c$

θ) $I=\int e^{2x}\sin x\,dx = \sin x\,\frac{e^{2x}}{2} - \int \frac{e^{2x}}{2}\cos x\,dx = \sin x\frac{e^{2x}}{2} - \frac{1}{2}\left(\cos x\frac{e^{2x}}{2} - \int \frac{e^{2x}}{2}\right.$

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γ) $I = \int \tan^{-1} 2x\,dx = \int \frac{\tan^{-1} t}{2}\,dt = \frac{1}{2}\int \tan^{-1} t\,dt = \frac{1}{2}\left[\tan^{-1} t \cdot t - \int t \cdot \frac{1}{1+t^2}\,dt\right]$

$t = 2x$

$dt = 2dx$

$dx = \frac{dt}{2}$ $\quad v = 1+t^2$

$dv = 2t\,dt$

$= \frac{1}{2}\left[\tan^{-1} t \cdot t - \int \frac{1}{2v}\,dv\right] = \frac{1}{2}\left[\tan^{-1} t \cdot t - \frac{1}{2}\ln|v|\right]$

$= x\tan^{-1} 2x - \frac{1}{4}\ln|1+4x^2| + c$

5.6

① β) $\int \frac{1}{x^2\sqrt{16-x^2}}\,dx = \int \frac{\cos v}{4\sin^2 v\sqrt{16-16\sin^2 v}}\,dv = \int \frac{\cos v}{4\sin^2 v\sqrt{16\cos^2 v}}\,dv$

$= \frac{1}{16}\int \frac{1}{\sin^2 v}\,dv = \frac{1}{16}\int \csc^2 v\,dv = -\frac{\cot v}{16} = -\frac{\sqrt{16-x^2}}{16x} + c$

$x = 4\sin v$

$v = \arcsin\frac{x}{4}\,dx$

$dx = 4\cos v\,dv$

ε) $I = \int \frac{2x^2}{\sqrt{9+x^2}}\,dx = 2\int \frac{x^2}{\sqrt{9+x^2}}\,dx = \int \frac{27\sec^2 v\tan^2 v}{\sqrt{9\tan^2 v + 9}}\,dv$

$= 9\int \sec v\tan^2 v\,dv = 9\int \sec v(\sec^2 v - 1)\,dv$

$= 9\int (\sec^3 v - \sec v)\,dv = \frac{\sec v\tan v}{2} + \frac{1}{2}\int \sec v\,dv$

$= \frac{\ln|\tan v + \sec v|}{2} + \frac{\sec v\tan v}{2}$

$x = 3\tan v$

$v = \arctan\left(\frac{x}{3}\right)$

$dx = 3\sec^2 v\,dv$

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γ) $I = \int \tan^{-1} 2x\,dx = \int \frac{\tan^{-1} t}{2} dt = \frac{1}{2}\int \tan^{-1} t\,dt = \frac{1}{2}\left[\tan^{-1} t \cdot t - \int t \cdot \frac{1}{1+t^2} dt\right]$

$t = 2x$

$dt = 2dx$

$dx = \frac{dt}{2}$ $\quad u = 1+t^2$

$du = 2t\,dt$

$= \frac{1}{2}\left[\tan^{-1} t \cdot t - \int \frac{1}{2u} du\right] = \frac{1}{2}\left[\tan^{-1} t \cdot t - \frac{1}{2}\ln|u|\right]$

$= x\tan^{-1} 2x - \frac{1}{4}\ln|1+4x^2| + c$

5.6

① β) $\int \frac{1}{x^2\sqrt{16-x^2}} dx = \int \frac{\cos u}{4\sin^2 u \sqrt{16-16\sin^2 u}} du = \int \frac{\cos u}{4\sin^2 u \sqrt{16\cos^2 u}} du$

$= \frac{1}{16}\int \frac{1}{\sin^2 u} du = \frac{1}{16}\int \csc^2 u\,du = -\frac{\cot u}{16} = \frac{\sqrt{16-x^2}}{16x} + c$

$x = 4\sin u$

$u = \arcsin\frac{x}{4}\,dx$

$dx = 4\cos u\,du$

ε) $I = \int \frac{2x^2}{\sqrt{9+x^2}} dx = 2\int \frac{x^2}{\sqrt{9+x^2}} dx = \int \frac{27\sec^2 u \tan^2 u}{\sqrt{9\tan^2 u + 9}} du$

$= 9\int \sec u \tan^2 u\,du = 9\int \sec u(\sec^2 u - 1) du$

$= 9\int (\sec^3 u - \sec u) du = \frac{\sec u \tan u}{2} + \frac{1}{2}\int \sec u\,du$

$= \frac{\ln|\tan u + \sec u|}{2} + \frac{\sec u \tan u}{2}$

$x = 3\tan u$

$u = \arctan\left(\frac{x}{3}\right)$

$dx = 3\sec^2 u\,du$

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6.1

6.① β) $I=\int_{-1}^{2} x(1+x^3)dx = \int_{-1}^{2}(x+x^4)dx = \left[\frac{x^2}{2}+\frac{x^5}{5}\right]_{-1}^{2} = \frac{81}{10}$ τμ.

③ α) $I=\int_{-1}^{1} x^2(x^3+2)^{-3}dx = -\frac{1}{6}\cdot\left[(x^3+2)^{-2}\right]_{-1}^{1} = \frac{4}{27}$ τμ.

δ) $I=\int_{1}^{2}(x^2+1)\ln x\,dx = \int_{1}^{2}(x^2\ln x+\ln x)dx$

$= \int_{1}^{2} x^2\ln x\,dx + \int_{1}^{2}\ln x\,dx.$

$= \frac{x^3}{3}\ln x - \int_{1}^{2}\frac{x^3}{3}\cdot\frac{1}{x}\cdot dx + x\ln x - \int_{1}^{2} x\cdot\frac{1}{x}dx$

$= \left[\frac{x^3}{3}\ln x - \frac{x^3}{9} + x\ln x - x\right]_1^2 = \frac{14}{3}\ln 2 - \frac{16}{9}$

6.2

①. α) $\int_{1}^{6}(9x-x^2)dx = \left[\frac{9x^2}{2}-\frac{x^3}{3}\right]_1^6 = \frac{516}{6}$ τμ.

β) $\int_{0}^{\pi} x\cos\frac{x}{2}dx = \int_{0}^{\pi} x\,d\left(2\sin\frac{x}{2}\right) = 2x\sin\frac{x}{2} - 2\int_{0}^{\pi}\sin\frac{x}{2}dx.$

$= \left[2x\sin\frac{x}{2}+4\cos\frac{x}{2}\right]_0^{\pi} = (2\pi-4)$ τμ.

② β) $\int_{0}^{2}(9x-x^2)dx = \left[\frac{9x^2}{2}-\frac{x^3}{3}\right]_0^2 = \frac{46}{3}$ τμ.

ΠΡΩΤΟΠΟΡΙΑ ΤΗΣ ΠΑΤΡΙΑΣ ΟΥΚ ΕΛΑΤΤΩ ΠΑΡΑΔΟΣΟΦ ΤΕΧΝΟΛΟΓΙΚΟΥ ΠΑΝΕΠΙΣΤΗΜΙΟΥ ΚΥΠΡΟΥ

6.3

③ a) $\frac{\int_0^{1/100} 400\sin\pi t\,dt}{\frac{1}{100}-0} = 400\int_0^{1/100}\sin(\pi t)\,dt = \frac{-400\cos\left(\frac{\pi}{100}\right)+400}{\frac{\pi}{100}}$

β) R.M.S $= \frac{\int_0^{1/100}\sin^2\pi t\,dt}{\frac{1}{100}} = \frac{800\pi - 4000\sin\left(\frac{\pi}{50}\right)}{100\pi}$

$RMS = \sqrt{\frac{800\pi - 4000\sin\frac{\pi}{50}}{100\pi}} = \sqrt{0,52\ldots} = 0,72$

⑦

$MV = \frac{\frac{\pi^3}{2} + \frac{2\pi(-1)}{2}}{5\pi} = \frac{\pi^2-2}{10}$.

RMS.

6.4

① α) $V = \pi\int_0^1 \left[(x+1)^2 - x^4\right]dx = \pi\int_0^1 (x^2+2x+1-x^4)\,dx$

$= \frac{32\pi}{15}$ κ.μ.

γ) $V = \pi\int_{-2}^{2}(4-x^2)^2\,dx = \frac{512\pi}{15}$ κ.μ.

③ γ) $V = \pi\int_0^4\left[\left(\frac{x}{3}\right)^2 - (4-x)^2\right]dx = \frac{512\pi}{27}$ κ.μ.

ΠΡΩΤΟΠΟΡΙΑ ΤΗΝ ΠΑΤΡΙΔΑ ΠΑΡΑΔΙΔΩΣΩ ΤΕΧΝΟΛΟΓΙΚΟΥ ΠΑΝΕΠΙΣΤΗΜΙΟΥ ΚΥΠΡΟΥ

1.2 $A_f \in \mathbb{R}$ $A_g \in [0,+\infty)$.

① δ)(i) $(f+g)(x) = 2x + \sqrt{x}$ $A_{f+g} [0,+\infty)$

(ii) $(f-g)(x) = 2x - \sqrt{x}$ $A_{f-g} [0,+\infty)$.

(iii) $(fg)(x) = 2x\sqrt{x}$ $A_{fg} [0,+\infty)$

(iv) $\left(\frac{f}{g}\right)(x) = \frac{2x}{\sqrt{x}}$ $A_{f/g} (0,+\infty)$

(v) $\left(\frac{g}{f}\right)(x) = \frac{\sqrt{x}}{2x}$ $A_{g/f} (0,+\infty)$

② γ) $A_f \in \mathbb{R}$. $A_g \in [1,+\infty)$

(i) $(f \circ g)(x) = f(g(x)) = \left(\sqrt{x-1}\right)^2 = x-1$

$A_{f\circ g} = \{x \in [1,+\infty) : \sqrt{x-1} \in \mathbb{R}\} = \{x \geq 1 \Rightarrow x \in [1,+\infty)$

(ii) $(f \circ f)(x) = f(f(x)) = (x^2)^2 = x^4$

$A_{f\circ f} = \{x \in \mathbb{R} : x^2 \in \mathbb{R}\} = \mathbb{R}$

(iii) $(g \circ f)(x) = g(f(x)) = \sqrt{x^2-1}$

$A_{g\circ f} = \{x \in \mathbb{R} : x^2 \in [1,+\infty)\} = \mathbb{R} \cdot [1,+\infty)$

(iv) $(g \circ g)(x) = g(g(x)) = \sqrt{\sqrt{x-1}-1}$

$A_{g\circ g} = \{x \in [1,+\infty) : \sqrt{x-1} \in [1,+\infty)\}$

$\left.\begin{array}{l} x \geq 1 \\ \sqrt{x-1} \geq 1 \\ x-1 \geq 1 \\ x \geq 2 \end{array}\right\} A_{g\circ g} \in [2,+\infty)$.

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